CAR BLACK BOX

System Operations & User Manual

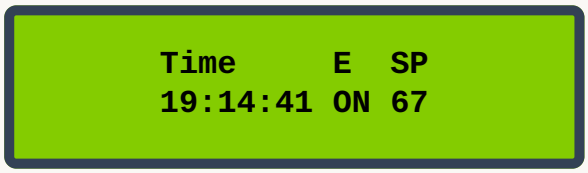
Final Specifications (Revised Switch Logic)

Includes updated Set Time logic, Password Mismatch delays, and dedicated Exit mappings.

1. System Boot & Dashboard State

Upon turning on the vehicle (powering the system), the system boots directly into the Dashboard mode. The initial default event logged is always **ON**.

Dashboard LCD Display Format



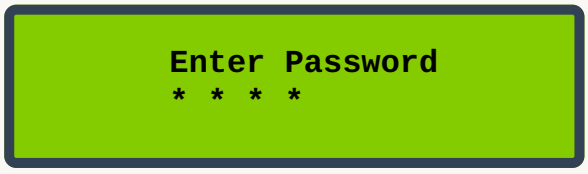
Speed Tracking: The external potentiometer controls the speed. The ADC signal (0-1023) is continuously mapped to a speed value from **0 to 99**.

Dashboard Controls

Switch	Action	System Response
SW1	Collision	Registers a collision. The Event updates to **C** and is logged.
SW2	Gear Up	Increments gear (e.g., **GN → G1 → G2 → G3 → G4**).
SW3	Gear Down	Decrements gear (e.g., **G4 → G3 → G2 → G1 → GN → GR**).
SW4	Menu Access	Long press to open the Security Login Screen.

2. Security Login Screen

Press **SW4** from the Dashboard to access settings. You must enter a 4-digit binary password.



- **SW4:** Enters a **1**.

- **SW5:** Enters a **0**.

Inactivity Timeout: If no switch is pressed for 5 seconds, the system automatically exits to the Dashboard.

Failed Attempts: You get 3 attempts. If all fail, the system is blocked for a predefined time (e.g., 2 minutes) showing: `wait for ..... 2 min`

3. Main Menu Navigation

The scrolling menu allows access to logs and settings. An asterisk (*****) indicates the highlighted option.

```
*View log
 Clear log
```

Main Menu Controls

Switch	Function in Main Menu
SW1	**Scroll Up** the menu list.
SW2	**Scroll Down** the menu list.
SW4	**Enter** the highlighted option.
SW5	**Exit to Dashboard.** (Universal Back button).

4. Specific Menu Operations

4.1 View Log

Displays up to 10 stored event logs (Indexed 0-9).

```
# Time     E  SP
0 19:45:38 ON 00
```

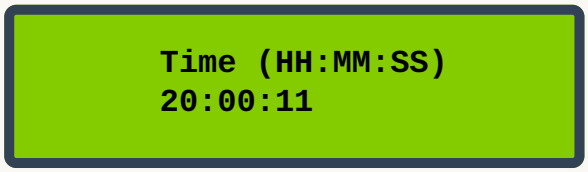
- **SW1:** Scroll Up.
- **SW2:** Scroll Down. *(Rule: If you reach the last log, e.g., Log #9, pressing SW2 will do nothing. The screen will not wrap around).*
- **SW5:** Exit back to the menu/dashboard.

4.2 Clear Log & 4.3 Download Log

- **Clear Log:** Deletes all logs and displays `Logs cleared successfully`.
- **Download Log:** Transmits EEPROM data to a PC via UART (Tera Term).

4.4 Set Time

Allows adjustment of the Real-Time Clock. The selected field (HH, MM, or SS) blinks continuously.



Set Time Controls:

- **SW1: Increment** the value of the blinking field.
- **SW2: Change Field** (Moves cursor from Hours → Minutes → Seconds).
- **SW4: Save & Exit** (Saves the time and jumps directly back to the Dashboard).
- **SW5: Cancel & Exit** (Discards changes and returns to the Main Menu).

4.5 Change Password

Allows overwriting the 4-digit EEPROM password.

1. Enter the new password (SW4 for '1', SW5 for '0').
2. Re-enter the new password to verify.

Outcome Logic:

- **If Matched:** LCD shows `pw changes successfully` and returns to the Main Menu.
- **If Not Matched:** LCD shows `pass is not matching`. The system will **wait for exactly 2 seconds** and then automatically return to the Main Menu without saving.